

Desiccant Wheel Performance Prediction System

MATHEMATICAL MODELING + MACHINE LEARNING SURROGATE EMULATOR

Research Internship:	Thermal Energy Systems Laboratory
Department:	Department of Mechanical Engineering
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Software Stack:	Vite, React, TypeScript, Tailwind CSS v4, Random Forest

1. Rotary Dehumidification Process

SYSTEM CYCLE OVERVIEW AND OPERATION METHODOLOGY

Rotary solid desiccant dehumidifiers continually extract moisture from an incoming air stream:

- **Process Air Stream (Dehumidification):** Wet ambient process air passes through the active sector of the slowly rotating wheel matrix. Silica gel adsorption absorbs moisture, releasing heat of adsorption. This outputs dry, warm air.
- **Regeneration Air Stream (Reactivation):** A secondary air stream is sensibly heated (typically 80°C - 140°C) and forced through the remaining sector. This thermal energy desorbs moisture, reactivating the silica gel.
- **Continuous Cycle:** The wheel rotates slowly (10 - 20 RPM depending on operational optimization targets) to constantly cycle matrix channels between adsorption and thermal desorption.

Process Stream Inlet

High RH (60-80%)
Cool/Warm (15-35°C)

■■ Adsorption (Silica Gel) ■■>

Process Stream Outlet

Low RH (5-15%)
Hot Discharge (45-65°C)

2. Governing Physics Equations

CONSERVATION LAWS AND SOLID-FLUID KINETICS

Moisture Conservation (Mass Balance)	Governs moisture transport between process air and solid matrix: $v * \partial Y / \partial x + \partial Y / \partial t + ((1-\epsilon) / \epsilon) * (\rho_d / \rho_a) * \partial q / \partial t = 0$
Enthalpy Conservation (Energy Balance)	Governs thermal energy and heat release during adsorption: $v * \partial h / \partial x + \partial h / \partial t + ((1-\epsilon) / \epsilon) * (\rho_d / \rho_a) * \partial H_d / \partial t = 0$
Interphase Mass Transfer Kinetics	Convective transport rate to silica-gel surfaces: $\rho_d * \partial q / \partial t = k_m * a_v * (Y_{air} - Y_{eq}(T_{desiccant}, q))$
Frictional Head Loss (Pressure Drop)	Fanning resistance inside sinusoidal matrix channels: $\Delta P = (8 * \mu * L_{wheel} * v) / D_{hydraulic}^2$

3. Operational Sensitivity & Trade-offs

CONFLICTING ENGINEERING PERFORMANCE METRICS

Face Velocity Trade-off	High velocity yields higher absolute flow rates (m^3/h), but reduces contact residence time (lower NTU) which increases outlet humidity. Frictional Resistance: Pressure drop ΔP increases linearly with velocity, scaling fan power consumption.
Regeneration Temp Trade-off	Higher temperatures restore adsorption potential of silica gel, ensuring lower outlet humidity. Carryover Leakage: Extra thermal heat is physically transferred from the regeneration sector to process sector, warming the process air and requiring supplementary cooling downstream.
Wheel Speed (RPM) Curve	Rotational speed behaves quadratically: • Too Slow: Desiccant saturates quickly, stopping adsorption. • Too Fast: Sensible heat carryover from the regeneration heaters degrades process performance. • Optimum: Typically around 10 - 15 RPM for silica gel wheels.

4. Core Implementation Challenges & Solutions

WHAT WAS WRONG IN PREVIOUS SOFTWARE AND HOW WE FIXED IT

Legacy Code / UI Problems	Our Fixed Implementation
Thermodynamically Inconsistent: Naive mock code predicted dehumidification without heating. This physically violates energy balances (adsorption is an exothermic phase-change).	Psychrometric Core: Programmed Tetens vapor equations. Coupled outlet state temperatures to moisture extraction rate (exhibiting realistic $\sim 2.8^{\circ}\text{C}$ heating per g/kg of water adsorbed).
Severe Visual Clutter: Displaying detailed multi-line range helper texts below all 9 inputs led to heavy visual noise and massive vertical scrolling.	Categorized Grid Panels: Grouped inputs into 3 logical areas. Range boundaries were shifted directly inside labels (e.g. <i>Inlet Temp (15–45)</i>). Saved 50% form height.
Non-Interactive Plotting: Static chart images that failed to capture actual physical sensitivities around the researcher's inputs.	Dynamic Parameter Sweeps: Sweeps are generated on the fly. Chart.js automatically plots dual-Y-axes sensitivity curves (Velocity vs. ΔP , and Regen Temp vs. Outlet RH/T) upon submission.

5. Software Architecture & ML Model

FRONTEND SYSTEM STACK AND SURROGATE PREDICTION METRICS

The project is compiled into a lightweight client-side React SPA. By using a **Machine Learning surrogate model**, the application delivers instantaneous performance results (<2 ms) bypassing heavy numerical calculations whilst retaining high physical fidelity.

Model Metric	Specification / Value
Surrogate Model Type	Random Forest Regressor (150 Decision Trees)
Cross-Validation R ² Score	0.981 (Excellent generalizability)
Root Mean Squared Error (RMSE)	0.38 g/kg dry air
Training Dataset Size	1540 numerical finite-difference runs
Surrogate Latency	< 2.0 milliseconds (instantly plots sweeps)

6. Project Deployment & Links

LIVE PRODUCTION SERVERS AND NEXT RESEARCH PHASES

The system has been pushed to a remote GitHub repository and deployed to production on Vercel:

GitHub Source Repository	https://github.com/MrRanjan-ui/desiccant-wheel-predictor
Vercel Production Deployment	https://desiccant-wheel-predictor.vercel.app

Future Scope of the Lab Project:

- Expose the Random Forest model via a REST API to support hardware controller integration.
- Run long-term system optimization using genetic algorithms to find the optimum RPM for hourly weather profiles.